



UPES ACM & ACM-W

100 Days of Code

Day: 12

Problem 1

Number of Lines To Write String (LeetCode #806)

<https://leetcode.com/problems/number-of-lines-to-write-string/description/>

Description

A problem that focuses on calculating how many lines are required to write a string when each character occupies a specified width.

Given an array widths representing the width of each lowercase English letter and a string s, determine the total number of lines needed and the width occupied by the last line. Each line can contain at most 100 units.

This problem helps build concepts like:

- String traversal
- Character width calculation
- Basic simulation
- Implementation skills

These concepts are useful for solving implementation-based string problems.

Your Task: Calculate the number of lines required to write the string using an optimized approach.

Input

An array widths of size 26 representing each letter's width, and a string s.

Output

Print an array of two integers: total lines needed and width of the last line.

Sample Input 1	Sample Output 1
Widths=[10,10,10...] , s = "abcdefghijklmnopqrstuvwxyz"	[3,60]
Sample Input 2	Sample Output 2
Widths=[10,10,10...] , s= "bbbcccdadaa"	[2,4]

Problem 2

Smallest Distinct Window

Description

A problem that focuses on finding the shortest substring containing every distinct character present in the given string.

Given a string str, return the length of the smallest substring that contains each distinct character of the string at least once.

This problem helps build concepts like:

- String traversal
- Substring processing
- Character comparison
- Basic string manipulation

These concepts are useful for solving a variety of string-based problems.

Your Task: Find the length of the smallest valid substring using an optimized approach

Input

A string str.

Output

Print the length of the smallest substring containing all distinct characters..

Sample Input 1	Sample Output 1
Str= "abcda"	4
Sample Input 2	Sample Output 2
Str= "zzxyzz"	3
Sample Input 3	Sample Output 3
Str= "abcaac"	3